FUNMILOLA AKEJU

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PROFESSIONAL SUMMARY

Electrical and Computer Engineering Ph.D. candidate with over 8 years of telecommunications and IT experience. Strong research background in signal processing, specializing in channel estimation, signal detection, and equalization algorithms for MIMO-OFDM and NOMA systems. Awarded for excellence in wireless communications research.

EDUCATION

Morgan State University, Baltimore, MD, USA
Ph.D. in Electrical and Computer Engineering | GPA: 3.86 | 2023 – Present

National Open University of Nigeria
M.Sc. Information Technology | 2023

Olabisi Onabanjo University, Nigeria
B.Sc. Computer Engineering | 2013

SKILLS SUMMARY

Languages: MATLAB, Python, C++, SQL
Frameworks: TensorFlow, Keras, PyTorch, Scikit-Learn, Pandas, NumPy
Tools: Power BI, Microsoft Excel, Asterisk (Elastix IP/PBX), Office 365 Admin
Platforms: Windows OS, Mac OS, WordPress
Soft Skills: Research, Technical Documentation, Team Management

WORK EXPERIENCE

Graduate Research Assistant | Morgan State University | Jan 2023 – Present

- Implement LS channel estimation and signal detection algorithms for NOMA MIMO-OFDM systems
- Apply SIC for channel estimation and equalization
- Investigate Physics-Informed Machine Learning (PIML) for communication systems
- Design simulations in MATLAB and Python neural network models

Information Technology Analyst | Greenwich Trustees Limited | Feb 2020 – Sep 2022

- Managed enterprise IT infrastructure and automated workflows using Power BI
- Coordinated MyTrade Book mobile app development
- Implemented ITILv4 frameworks for system optimization
- Managed secure remote access and web assets

IT Support Specialist | Meyer Plc | Apr 2018 – Jan 2020

- Implemented Elastix VoIP systems and network security controls
- Maintained perimeter security and resolved vulnerabilities

PROJECT EXPERIENCE

Wireless Signal Recovery (Doctoral Research)

- CNN-based signal recovery for wireless systems
- MATLAB and Python implementation for performance optimization

Advanced Machine Learning & Predictive Modeling

- RNN, LSTM, GRU for time-series forecasting (crypto)
- CNN and MLP for classification tasks
- Regression models for environmental and industrial datasets
- Tech stack: PyTorch, Keras, Scikit-learn, Jupyter

PUBLICATIONS & AWARDS

- SPIE 2026: Improved SIC Signal Detection for OFDM NOMA (accepted)
- IEEE CISS 2025: LS Channel Estimation with SIC Equalization
- ITC 2024: Channel Estimation for MIMO-OFDM Systems
- Outstanding Research Award (2nd Place), ITC 2024

CERTIFICATIONS

- ISC2 Cybersecurity Certification, 2025
- CCNA, Charles Sturt University, 2020
- Computer Communications, Saylor Academy, 2020
- Sage ERP 100 Technical Training, 2018